

Broiler farm documentation

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2. Subsystem Use Cases

A. UC-13: Receiving Delivery Notice from Management

Use Case ID: UC-13

Use Case Name: Receiving Delivery Notice from Management

Actor: Slaughterhouse Manager

Preconditions:

- DeliveryNotice must be sent before this scenario will happen.
- Scheduled delivery date/time set.
- System integration with Slaughterhouse active.

Main Flow:

- The Manager will confirm the receipt of this note from the Management subsystem
- System receives the following data package from the Management.
 - Delivery notice ID
 - Scheduled arrival date/time
 - Lot information (ID, age, breed)
 - Estimated quantity and weight
 - Average bird weight
 - Quality grade
 - Health status
 - Treatment history summary
 - Withdrawal period confirmations
 - Special handling instructions
 - Transport details
 - Contact information
- System updates DeliveryNotice status to Received
- System records receipt timestamp

Alternative Flows:

- **A1: Missing Critical Data**
 - Returns to Management System
 - System lists missing fields
- **A2: Slaughterhouse Rejects Delivery**
 - System receives rejection message with reason
 - System notifies Farm Manager
 - System suggests alternative dates
 - Farm Manager must reschedule

Postconditions:

- DeliveryNotice received successfully
- Status updated to RECEIVED
- SlaughterLot will be created and linked to this DeliveryNotice

Business Rules:

- All Notices must be send at least 24 hours before scheduled delivery
- After receiving the Notice, the Slaughterhouse manager must create the Slaughter Lot document

Integration Points:

- **Inbound from Management:**
 - Delivery schedule
 - Lot details
 - Quality metrics
 - Health history
 - Treatment records

B. UC-14: Create a Slaughter Lot

Use Case ID: UC-14

Use Case Name: Create a Slaughter Lot

Actor: Slaughterhouse Manager

Preconditions:

- A valid **Delivery Notice** must be received and marked as Received.
- Chickens must be physically delivered to the Slaughterhouse.

- The Delivery Notice must include lot details (ID, breed, age, weight, etc.).

Main Flow:

- The Manager accesses the Create Slaughter Lot module.
- The system retrieves data from the associated Delivery Notice.
- The Manager reviews and confirms the following data:
 1. Delivery Notice ID
 2. Supplier farm and transport details
 3. Arrival timestamp and lot identifiers
- Quantity and total weight of chickens
 - The Manager validates or adjusts the values if needed.
 - The system creates a new **Slaughter Lot**, automatically linking it to the Delivery Notice.
 - System updates the Lot status to Ready for Inspection.
 - Timestamp of creation is recorded.

Alternative Flows:

A1: Missing Data:

- System notifies the user of incomplete fields.
- Manager must correct data before confirming.

A2: Delivery Rejected:

- Manager can cancel the creation.
- System records reason for cancellation.

Postconditions:

- A new Slaughter Lot is created and linked to its Delivery Notice.
- The lot is now available for Ante-Mortem Inspection.

Business Rules:

- Each Delivery Notice can generate only one Slaughter Lot.
- A Slaughter Lot cannot proceed to processing without completed inspection.

C. UC-15: Create Ante-Mortem Inspection Sheet

Use Case ID:

Use Case Name: Create Ante-Mortem Inspection Sheet

Actor: Slaughterhouse Manager

Preconditions:

- A Slaughter Lot must exist and be marked as Ready for Inspection.
- The Vet must be logged in with proper access rights.

Main Flow:

- The Vet opens the Ante-Mortem Inspection Sheet creation form.
- The system retrieves the linked Slaughter Lot data.
- The Vet enters or confirms the following:
 - Lot ID, date, and time of inspection
 - Visual condition of birds
 - Mortality rate upon arrival
 - General health observations
 - Temperature, behavior, and physical anomalies
 - The Vet saves the Inspection Sheet as Draft or Pending Approval.
 - System stores the sheet and links it to the respective lot.

Alternative Flows:**A1: Missing or inconsistent data:**

- System displays validation errors.
- Vet corrects and resubmits.

A2: Vet unavailable:

- System allows Quality Control Officer to perform inspection temporarily.

Postconditions:

- Inspection Sheet created and linked to the Slaughter Lot
- Lot status updated to Awaiting Approval.

Business Rules:

- Ante-Mortem inspection must be completed before processing starts.
- Only authorized veterinary or QC staff can create the inspection sheet.

D. UC-16: Monitor Daily Stats**Use Case ID:**

Use Case Name: Monitor Daily Stats

Actor: Slaughterhouse Manager

Preconditions:

- Daily processing operations must be recorded in the system.
- All related sheets (Inspection, Slaughter, Packaging, Waste) must exist for that day.
- The Manager must have read-only or administrative access.

Main Flow:

- The Manager accesses the Daily Statistics dashboard.
- The system aggregates data from all relevant modules:
 - Total birds received and processed
 - Total carcasses rejected
 - Average yield percentage
 - Total packaged and delivered quantities
 - Waste and loss metrics
 - The Manager can apply filters (by date, lot, or production line).
 - The system displays charts and summaries for quick analysis.
 - Manager exports or prints daily reports as needed.

Alternative Flows:**A1: Missing data:**

- System notifies the Manager that some sheets are incomplete.
- Partial statistics are shown with a warning.

Postconditions:

- Daily operational metrics successfully displayed and optionally exported.

Business Rules:

- Daily stats must always match totals in Slaughter and Waste reports.
- Only Managers and authorized users can access cumulative data.

E. UC-17: Approve the Inspection Sheet**Use Case ID:**

Use Case Name: Approve the Inspection Sheet

Actor: Veteranian

Preconditions:

- The Ante-Mortem Inspection Sheet must exist in Pending Approval state.
- The Vet or QC Officer must have approval rights.

Main Flow:

- The Vet reviews the completed inspection data.

- The system displays all relevant fields: health status, mortality, anomalies, etc.
- The Vet clicks “Approve” to confirm the inspection.
- System updates the sheet status to Approved.
- The linked Slaughter Lot status changes to Ready for Processing.
- Timestamp and approver credentials are recorded.

Alternative Flows:

A1: Rejection:

- Vet clicks “Reject”.
- System requests reason for rejection.
- Status changes to Rejected, and the Manager is notified.

A2: Missing data:

- System blocks approval and highlights missing or invalid fields.

Postconditions:

- Inspection sheet status updated to Approved or Rejected.
- Slaughter Lot proceeds or halts based on decision.

Business Rules:

- Only certified veterinary staff may approve inspection sheets.
- Approval is irreversible unless a corrective action process is initiated.

F. UC-18: Create Delivery Notice

Use Case ID:

Use Case Name: Create Delivery Notice

Actor: Cold Storage Manager

Preconditions:

- Packaged products must exist in Cold Storage and be marked Ready for Dispatch.
- Client order or schedule must be registered in the system.

Main Flow:

- Logistics Manager opens the Create Delivery Notice interface.
- System retrieves available packaged lots.
- Manager selects batches and assigns them to a client order.
- System auto-generates a Delivery Notice with:
 - Notice ID, client details, delivery date
 - Quantity and total weight
 - Product types and packaging details
 - Manager reviews and confirms.
 - System records the Delivery Notice and updates stock accordingly.

Alternative Flows:**A1: Insufficient stock:**

- System alerts the user that requested quantity exceeds available stock.
- Manager must adjust selection.

Postconditions:

- New Delivery Notice created and linked to packaged lots.
- Stock reduced accordingly.

Business Rules:

- Only packaged lots with Approved quality status can be delivered.
- Delivery Notice must reference at least one Packaging Sheet.

G. UC-19: Processing Starts**Use Case ID:**

Use Case Name: Processing Starts

Actor: Workers

Preconditions:

- The Slaughter Lot must have an Approved Ante-Mortem Inspection Sheet.
- Production line and equipment must be ready and operational.

Main Flow:

- Supervisor opens the Processing Control Panel.
- System verifies all preconditions (lot approval, equipment readiness).
- Supervisor selects Start Processing.
- System changes the Slaughter Lot status to In Progress.
- Time of start is logged automatically.
- The operation data (input birds, line used) begins tracking.

Alternative Flows:**A1: Missing Inspection Approval:**

- System blocks start and displays error

A2: Equipment not ready:

- System delays start until all sensors confirm readiness.

Postconditions:

- Slaughter Lot marked as In Progress.
- Automatic logging of start time and initial parameters initiated.

Business Rules:

- Processing cannot begin without a valid, approved inspection.
 - System must record start events automatically.
-

H. UC-20: Processing finishes**Use Case ID:**

Use Case Name: Processing finishes

Actor: Workers

Preconditions:

- The Slaughter Lot must be in In Progress status.
- Processing operation must have been started and tracked.

Main Flow:

- The Supervisor monitors production through the system dashboard.
- When slaughtering completes, the Supervisor selects **Finish Processing**.
- The system automatically collects operational data:
 - Total number of birds processed
 - Total carcasses rejected
 - Average yield and total meat weight
 - Waste generated
 - System updates the Slaughter Lot status to Processed.
 - A Slaughter Report is automatically generated.
 - System records completion timestamp and user credentials.

Alternative Flows:**A1: System error or incomplete data:**

- System requests manual confirmation of final numbers.
- Supervisor enters missing fields before confirming completion.

Postconditions:

- Slaughter Lot marked as Processed.
- Slaughter Report automatically created and linked to the lot.

Business Rules:

- Processing cannot finish if automatic data collection fails validation.
 - Only Supervisors or authorized Managers can confirm completion
-

I. UC-21: Update Cold Storage Log

Use Case ID:

Use Case Name: Update Cold Storage Log

Actor: Cold Storage Manager

Preconditions:

- Packaged products or batches must exist and be ready for cold storage entry.
- The Cold Storage module must be active and linked to Packaging Sheets.

Main Flow:

- System automatically logs every new packaged batch entering storage.
- Each log entry includes:
 - Lot ID, Product Type, Packaging ID
 - Entry Date/Time, Quantity, and Temperature
 - When products are shipped, system automatically removes them from inventory.
 - Cold Storage Log is updated in real time.
 - Historical entries remain archived for traceability.

Alternative Flows:

A1: Sensor Failure or Manual Entry Needed:

- System allows authorized users to input data manually.
- Validation required by Manager.

Postconditions:

- Cold Storage Log reflects real-time stock and movements.
- Data synchronized with Delivery Notices and Waste Reports.

Business Rules:

- All stock movements must be time-stamped and traceable.
 - Only the system can update logs automatically. Manual changes require approval.
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J. UC-22: Create Waste Report

Use Case ID:

Use Case Name: Create Waste Report

Actor: Workers

Preconditions:

- The Slaughter Report must exist and include data on non-usable parts.
- Processing must be completed.

Main Flow:

- System extracts waste data from the Slaughter Report.
- A new Waste Report is generated with:
 - Lot ID, date, and total processed quantity
 - Weight and type of waste (biological, packaging, etc.)
 - Percentage of total waste vs. input quantity
 - QC Officer reviews and confirms the report.
 - System saves and marks the report as Approved.

Alternative Flows:**A1: Inconsistent Data:**

- QC Officer corrects data manually before approval

A2: System Failure:

- Report creation delayed until automatic sync completes.

Postconditions:

- Waste Report successfully created and linked to the processed lot.
- Waste data available for daily statistics and regulatory reporting.

Business Rules:

- Waste percentage must not exceed regulatory thresholds.
- QC approval mandatory for final report validation.

K. UC-23: Create Slaughter Report

Use Case ID:

Use Case Name: Create Slaughter Report

Actor: Workers

Preconditions:

- The Slaughter Lot must have status Processed.
- Data collection from processing lines must be complete.

Main Flow:

- System automatically compiles operational data from the processing phase.
- Generated Slaughter Report includes:
 - Total birds processed
 - Total weight before and after processing
 - Rejected carcasses and reasons
 - Average yield percentage

- Operator name and line used
- Manager reviews and validates the generated data.
- Report saved in the system and marked as Finalized.

Alternative Flows:

A1: Missing Production Data:

- Manager adds missing values manually.
- System recalculates totals before finalizing.

Postconditions:

- Slaughter Report created and linked to corresponding Lot.
- Data becomes available for stats, waste, and packaging modules.

Business Rules:

- Slaughter Report must be finalized before packaging begins.
 - Report serves as the main source for waste and yield analysis.
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3. Entities and the logic of them

Entity Definitions, relationships and business limitations

a) BaseEntity (Abstract)

- id: Long (PK)
- createdAt: DateTime
- createdBy: Employee
- updatedAt: DateTime
- updatedBy: Employee
- isActive: Boolean
- version: Integer (for optimistic locking)

b) Employee

- employeeId: Long (PK)
- firstName: String
- lastName: String
- email: String (unique)
- phone: String

- role: Enum (FARM_MANAGER, LOGISTICS_MANAGER, TRANSPORT_MANAGER, WORKER, VETERINARIAN)

- hireDate: Date

- isActive: Boolean

- farm: BroilerFarm (FK)

c) TreatmentSheet

- sheetId: Long (PK)

- date: Date

- poultryHouse: PoultryHouse (FK)

- lot: ChicksLot (FK)

- veterinarian: Employee (FK)

- createdBy: Employee (FK)

- diagnosis: String

- treatmentReason: String

- status: Enum (DRAFT, ACTIVE, COMPLETED)

Relationships:

- Has many TreatmentSheetLines

Business Rules:

- One treatment sheet per day per house
- Requires veterinarian approval
- Cannot delete if referenced by consumption

d) TreatmentSheetLine

- lineId: Long (PK)

- treatmentSheet: TreatmentSheet (FK)

- medication: Consumable (FK)

- dosage: Decimal

- dosageUnit: String

- administrationMethod: Enum (WATER, FEED, INJECTION, SPRAY)

- duration: Integer (days)

- startDate: Date

- endDate: Date
- withdrawalPeriod: Integer (days)
- quantityUsed: Decimal
- batchNumber: String

Calculated Fields:

- $\text{endDate} = \text{startDate} + \text{duration}$
- $\text{slaughterAllowedDate} = \text{endDate} + \text{withdrawalPeriod}$

Business Rules:

- Auto-creates ConsumptionSheet entries
 - Blocks delivery during withdrawal period
- e) ObservationSheetLine
- lineId: Long (PK)
 - observationSheet: ObservationSheet (FK)
 - metricType: Enum (WEIGHT, FCR, MORTALITY, HEALTH, BEHAVIOR)
 - metricValue: Decimal
 - notes: String

Auto-calculated Metrics:

- Starting bird count (from previous week or reception)
- Ending bird count (current - mortality)
- Total mortality count and percentage
- Average daily mortality
- Total feed consumed (sum from ConsumptionSheets)
- Feed Conversion Ratio: $\text{totalFeed} / \text{totalWeightGain}$

Manual Input Required:

- Sample weight measurements (random sampling)
- Health observations
- Behavioral notes
- Environmental conditions

Business Rules:

- Created weekly (configurable day)
- Previous observation required (except first week)
- Used to populate DeliveryNotice

f) DeliveryNotice

- noticeId: Long (PK)
- creationDate: Date
- scheduledDate: DateTime
- poultryHouse: PoultryHouse (FK)
- lot: ChicksLot (FK)
- destination: String (Slaughterhouse)
- transportManager: Employee (FK)
- vehicleInfo: String
- driverInfo: String
- approvalStatus: Enum (DRAFT, APPROVED, IN_TRANSIT, DELIVERED)
- approvedBy: Employee (FK)

Relationships:

- Has many DeliveryNoticeLines
- Creates Transport record

g) DeliveryNoticeLine

- lineId: Long (PK)
- deliveryNotice: DeliveryNotice (FK)
- estimatedQuantity: Integer
- averageWeight: Decimal (from observation)
- totalEstimatedWeight: Decimal (calculated)
- qualityGrade: Enum (A, B, C)
- specialInstructions: String
- loadingBay: String

Calculated Fields:

- $\text{totalEstimatedWeight} = \text{estimatedQuantity} * \text{averageWeight}$

Business Rules:

- Cannot create if in withdrawal period
- Requires minimum age (35-42 days typically)
- Requires farm manager approval
- Auto-populated from latest ObservationSheet

h) Slaughterhouse

- slaughterhouseId: Long (PK)
- name: String
- location: String
- capacity: Integer
- manager: Employee (FK)

Relationships:

- Has many SlaughterLots
- Receives ChickenReception records
- Generates SlaughterReports, Carcass, Packaging, ColdStorage, and WasteReports

i) SlaughterLot

- lotId: Long (PK)
- slaughterhouse: Slaughterhouse (FK)
- lotNumber: String
- receptionDate: Date
- processedDate: Date
- sourceLot: ChicksLot (FK)
- totalChickens: Integer

Relationships:

- Contains many Carcass records
- Linked to SlaughterReport
- Linked to ChickenReception

j) ChickenReception

- receptionId: Long (PK)
- slaughterhouse: Slaughterhouse (FK)

- deliveryDate: DateTime
- lot: ChicksLot (FK)
- deliveredBy: Transport (FK)
- numberOfChickens: Integer
- averageWeight: Decimal

Relationships:

- Linked to SlaughterLot
- Linked to Transport

Business Rules:

- Must have valid transport approval
- Must pass initial quality check

k) AnteMortemInspection

- inspectionId: Long (PK)
- slaughterLot: SlaughterLot (FK)
- inspectionDate: DateTime
- inspector: Employee (FK)
- healthStatus: Enum {PASS, FAIL}
- remarks: String

Business Rules:

- Must be completed before slaughtering
- Failing lot cannot proceed to SlaughterLot

l) SlaughterReport

- reportId: Long (PK)
- slaughterLot: SlaughterLot (FK)
- reportDate: DateTime
- totalProcessed: Integer
- averageWeight: Decimal

Relationships:

- Linked to multiple Carcass entries

- Generated per SlaughterLot

Calculated Fields:

- $\text{averageWeight} = \text{totalWeight} / \text{totalProcessed}$

m) Carcass

- carcassId: Long (PK)
- slaughterLot: SlaughterLot (FK)
- weight: Decimal
- qualityGrade: Enum {A, B, C}
- packagingStatus: Enum {PACKAGED, COLD_STORAGE, WASTE}

Business Rules:

- Weight measured post-slaughter
- Quality grade assigned per inspection

n) Packaging

- packagingId: Long (PK)
- slaughterLot: SlaughterLot (FK)
- packagingDate: DateTime
- operator: Employee (FK)
- totalPackages: Integer

Relationships:

- Contains many PackagingLine records
- Linked to ColdStorage

o) PackagingLine

- lineId: Long (PK)
- packaging: Packaging (FK)
- carcass: Carcass (FK)
- packageWeight: Decimal
- packageLabel: String

Calculated Fields:

- $\text{packageWeight} = \text{sum}(\text{carcass weights})$

Business Rules:

- Each carcass can only be in one package

p) ColdStorage

- storageId: Long (PK)
- slaughterhouse: Slaughterhouse (FK)
- location: String
- capacity: Integer

Relationships:

- Stores Carcass and Packaging
- linked to ColdStorageLog

q) ColdStorageLog

- logId: Long (PK)
- coldStorage: ColdStorage (FK)
- item: Carcass or Packaging (FK)
- dateStored: DateTime
- dateRemoved: DateTime
- temperature: Decimal

Business Rules:

- Temperature must be monitored
- FIFO (first-in, first-out) policy

r) WasteReport

- reportId: Long (PK)
- slaughterhouse: Slaughterhouse (FK)
- reportDate: DateTime
- totalWasteWeight: Decimal

Relationships:

- Has many WasteReportLine

Calculated Fields:

- totalWasteWeight = sum(WasteReportLine.weight)

s) WasteReportLine

- lineId: Long (PK)

- wasteReport: WasteReport (FK)
- carcass: Carcass (FK)
- weight: Decimal
- reason: String {DAMAGED, EXPIRED, UNSUITABLE}

Business Rules:

- Only carcasses failing quality go here
- t) Transport (for outbound shipments)

Attributes:

- transportId: Long (PK)
- slaughterhouse: Slaughterhouse (FK)
- vehicleInfo: String
- driverInfo: String
- departureDate: DateTime
- destination: String
- approvalStatus: Enum {DRAFT, APPROVED, IN_TRANSIT, DELIVERED}

Relationships:

- Transports Packaging or Carcass to clients

Business Rules:

- Must be approved before departure
- Cannot exceed vehicle capacity